99.500 Applied Mathematics for Engineering

Spring Semester AY2026

Bessel Ordinary Differential Equations

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Week 5 Outline

- Bessel ODEs and Bessel functions
- Diffusive reaction in a cylindrical shaped catalyst

Bessel ODEs

real number

Bessel ODE: $x^2 y'' + xy' + (x^2 - p^2)y = 0$

Solution: $y = c_1 J_p(x) + c_2 Y_p(x)$

$J_p(x)$: Bessel function of the first kind of order p and argument x

$$J_p(x) = \sum_{n=0}^{\infty} \frac{(-1)^n \left(\frac{1}{2}x\right)^{2n+p}}{n!\Gamma(n+p+1)}$$

Γ gamma function

$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$

$Y_p(x)$: Bessel function of the second kind of order p and argument x

$$Y_p(x) = \frac{\cos(\pi p) J_p(x) - J_{-p}(x)}{\sin(\pi p)}$$

if $n \in Z^+$

$\Gamma(n) = (n-1)!$

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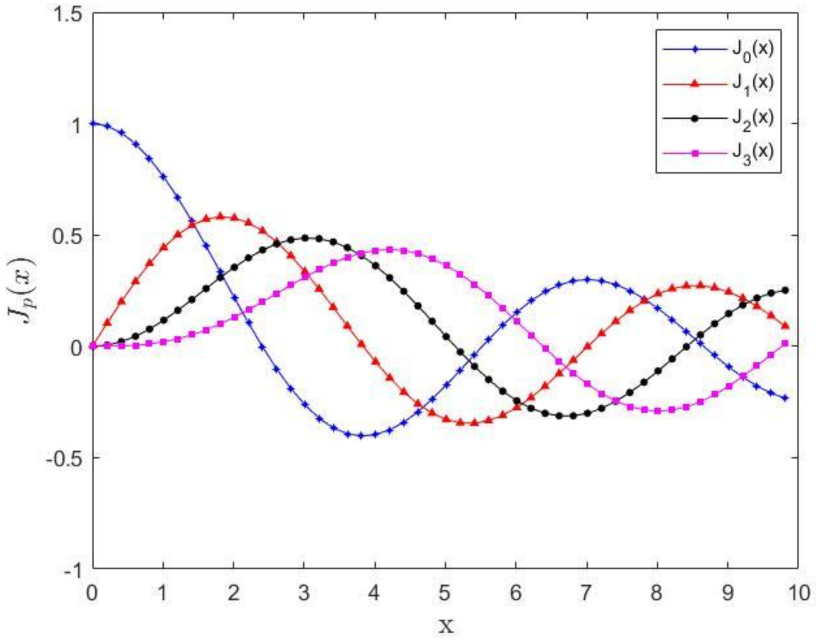
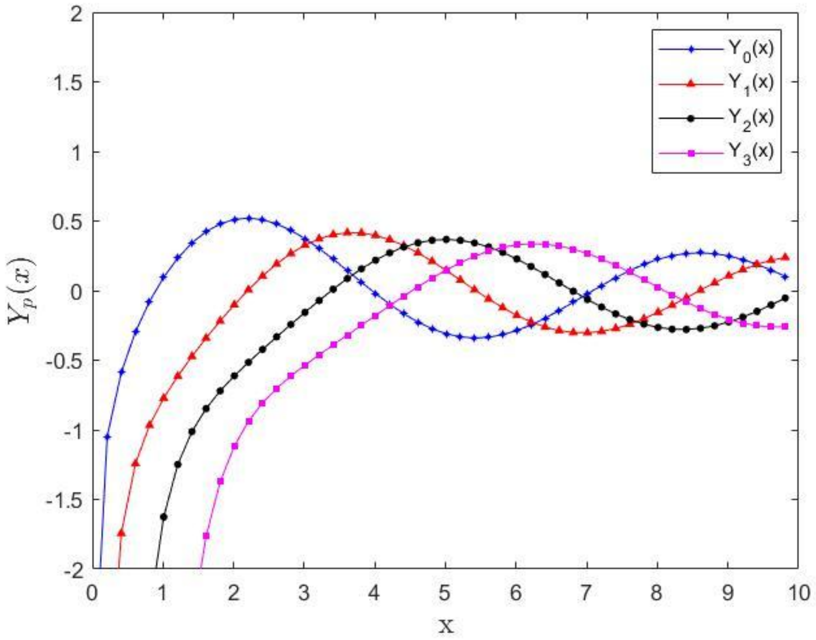
Bessel Functions

Matlab: besselj(p,x)



Matlab: bessely(p,x)



Modified Bessel ODEs

Modified Bessel ODE: $x^2y'' + xy' - (x^2 + p^2)y = 0$

Solution: $$y = c_1 I_p(x) + c_2 K_p(x)$$

$I_p(x)$: Modified Bessel function of the first kind of order p and argument x

$$I_p(x) = \sum_{n=0}^{\infty} \frac{\left(\frac{1}{2}x\right)^{2n+p}}{n!\Gamma(n+p+1)} \overset{?}{=} i^{-p} J_p(ix)$$

1

$$(-1)^n \cdot (-1)^n = (-1)^{2n}$$

$K_p(x)$: Modified Bessel function of the second kind of order p and argument x

$$K_p(x) = \frac{\pi}{2}\left[\frac{I_{-p}(x) - I_p(x)}{\sin(\pi p)}\right]$$

$$(-1)^n \cdot i^{2n}$$

Where does Bessel ODEs arise?

$$i^{-p} J_p(ix) = \sum_{n=0}^{\infty} \frac{(-1)^n \left(\frac{1}{2} i x\right)^{2n+p}}{n!\Gamma(n+p+1)} \cdot i^{-p} = \sum_{n=0}^{\infty} \frac{\left(\frac{1}{2}x\right)^{2n+p}}{n!\Gamma(n+p+1)} \cancel{\boxed{(-1)^n i^{2n+p} \cdot i^{-p}}}$$

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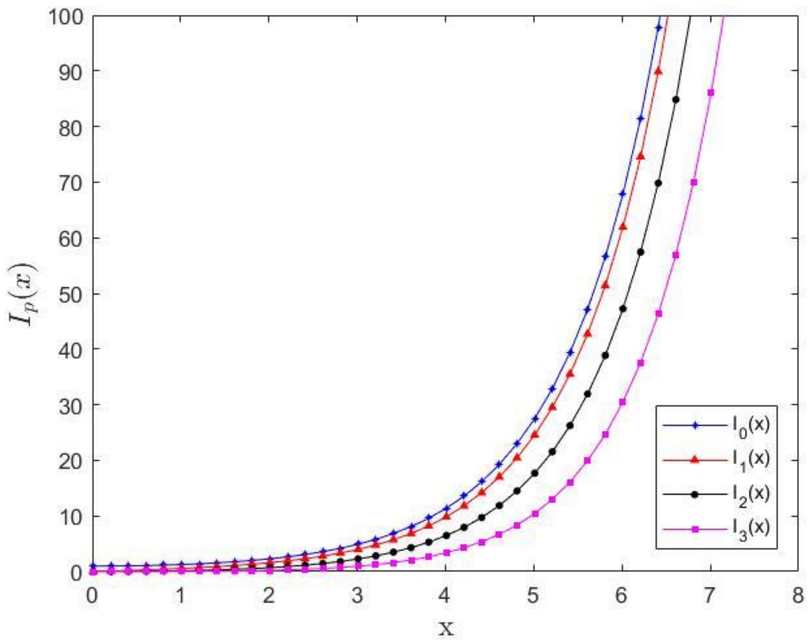
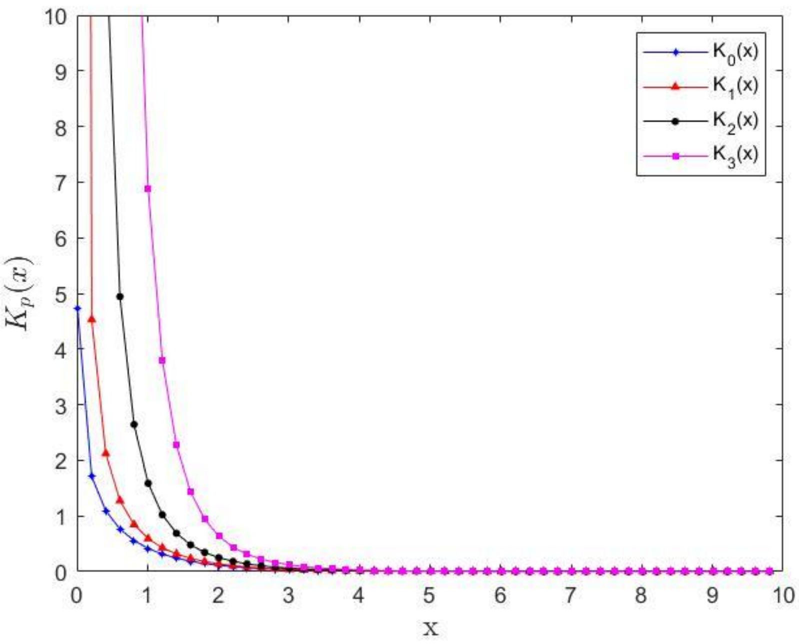
Bessel Functions

Matlab: besseli(p,x)



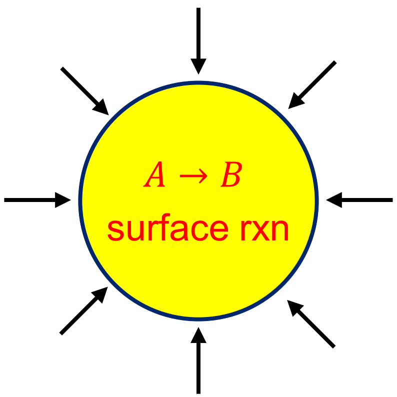
Matlab: besselk(p,x)

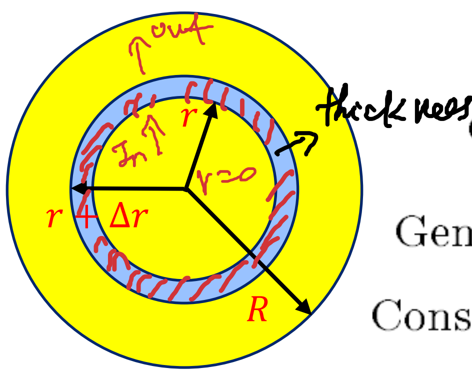


Problem generating Bessel ODEs

Bessel ODEs naturally arise in problem with radial symmetry

Example: Diffusive reaction in a cylindrical shaped catalyst





surface area of cylinder

In: $\left(-D\dfrac{dC_A}{dr}\underline{2\pi r L}\right)\Big|_r$

Out: $\left(-D\dfrac{dC_A}{dr}2\pi r L\right)\Big|_{r+\Delta r}$

Generation: 0

$\frac{m^2}{m^3} \cdot m^3 = m^2$

Consumption: $-\boxed{r_A}a\Delta V = -\boxed{kC_A}a(2\pi r\Delta r L)$

Specific surface area: $a\left(\dfrac{\text{m}^2}{\text{m}^3}\right)$ for catalyst

Accumulation: 0 (assuming steady-state)

Shell balance

In — Out — Consumption

$$\underbrace{\left(-\cancel{D}\frac{dC_A}{dr}\cancel{2\pi} r \cancel{L}\right)}_{\Delta r}\Big|_r - \underbrace{\left(-\cancel{D}\frac{dC_A}{dr}\cancel{2\pi} r\cancel{L}\right)}_{\Delta r}\Big|_{r+\Delta r} - \underbrace{kC_A a}_{D}(\cancel{2\pi} r\cancel{\Delta r}\cancel{L}) = 0$$

BC: $\begin{cases} \left.\dfrac{dC_A}{dr}\right|_{r=0} = 0 & \text{Symmetric} \\ \left(-D\dfrac{dC_A}{dr}2\pi r L\right)\Big|_{r=R} = \left[k_c\left(C_A - C_{A,0}\right)2\pi r L\right]_{r=R} \end{cases}$

diffusion of rxn — mass transfer (FYI)

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dC_A}{dr}\right) - \frac{ka}{D}C_A = 0 \overset{?}{\longrightarrow} x^2y'' + xy' - (x^2+p^2)y = 0$$

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Nondimensionalization

$\left[\dfrac{\left.\frac{dC_A}{dr}\cdot r\right|_{r+\Delta r} - \left.\frac{dC_A}{dr}\cdot r\right|_{r}}{\Delta r} - \dfrac{ka}{D}\cdot r C_A = 0\right]$ $\lim_{\Delta r \to 0}$

$\dfrac{d}{dr}\left(\dfrac{dC_A}{dr}\cdot r\right) - \dfrac{ka}{D}\cdot r C_A = 0$

$C_A = y\, C_{A,0}$ $\quad r = L_d x$

$$y = \frac{C_A}{C_{A,0}} \qquad x = \sqrt{\frac{ka}{D}}\, r = \frac{r}{L_d} \quad \text{where dynamic length } L_d = \sqrt{\frac{D}{ka}}$$

$= \dfrac{r}{\sqrt{\frac{D}{ka}}}$

divide by r to obtain (*)

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dC_A}{dr}\right) - \frac{ka}{D}C_A = 0 \implies \frac{1}{L_d x}\frac{d}{d(L_d x)}\left(L_d x \frac{d(C_{A,0}\,y)}{d(L_d x)}\right) - L_d^{-2}C_{A,0}\,y = 0$$

$$\implies x^2 y'' + xy' - x^2 y = 0 \text{ with } p = 0$$

$\dfrac{1}{x}\left[\dfrac{d}{dx}\left(x\cdot\dfrac{dy}{dx}\right)\right] - y = 0$

BC1: $$\left.\frac{dC_A}{dr}\right|_{r=0} = 0 \implies y'(0) = 0$$

$x^2\cdot\left[\dfrac{1}{x}\left(x\cdot y'' + y'\right) - y = 0\right]$

$x^2 y'' + xy' - x^2 y = 0$

BC2:

$$\left.\left(-D\frac{dC_A}{dr}\cancel{2\pi r L}\right)\right|_{r=R} = \left[k_c\left(C_A - C_{A,0}\right)\cancel{2\pi r L}\right]_{r=R}$$

$C_{A,0}\,y$ $\quad dr = dx\, L_d$ $\quad C_{A,0}\,y$

$$\implies -y'(\alpha) = \beta\left[(y(\alpha) - 1)\right]$$

$-D\dfrac{dy}{dx}\dfrac{\cancel{C_{A,0}}}{L_d} = k_c\,\cancel{C_{A,0}}(y-1)$

$-y' = \boxed{\dfrac{k_c\cdot L_d}{D}}(y-1)$

$= \beta$

$$\alpha = \frac{R}{L_d} = R\sqrt{\frac{ka}{D}} \;\&\; \beta = \frac{k_c}{\sqrt{kaD}}$$

when $r = R$. $\quad \alpha = \left.\dfrac{r}{L_d}\right| = \dfrac{R}{L_d}$.

$= \dfrac{k_c\cdot\sqrt{\frac{D}{ka}}}{D} = \dfrac{k_c}{\sqrt{kaD}}$

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Solution

$$x^2y'' + xy' - x^2y = 0 \qquad \text{Modified Bessel ODE with } p = 0$$

$$y(x) = c_1I_0(x) + c_2K_0(x) \xRightarrow{\text{diff}} \frac{d}{dx}\left[C_1 I_0(x) + C_2 K_0(x)\right] = C_1 I_1(x) - C_2 K_1(x)$$

$$\text{BC1. } y'(0) = 0 \implies c_1I_1(0) - \cancel{c_2\underbrace{K_1(0)}_{\infty}} = 0$$

$$\implies c_2 = 0 \;\&\; y(x) = c_1I_0(x)$$

$$\text{BC2 } -y'(\alpha) = \beta\left[(y(\alpha) - 1)\right] \implies -c_1I_1(\alpha) = \beta\left[(c_1I_0(\alpha) - 1)\right]$$

$$\implies c_1 = \frac{\beta}{\beta I_0(\alpha) + I_1(\alpha)}$$

$$\implies y(x) = \frac{\beta I_0(x)}{\beta I_0(\alpha) + I_1(\alpha)} \qquad \frac{\beta}{\beta I_0(\alpha) + I_1(\alpha)} = C_1$$

What happens if the derived model: $x^2y'' + xy' - \boxed{x^2y} = qx^2$ $\quad \boxed{x^2y} = x^2(z - q) = x^2 z - q x^2$

Let $z = y + q$ $\qquad x^2z'' + xz' - x^2z = 0$

$$z(x) = c_1I_0(x) + c_2K_0(x)$$

$$y(x) = c_1I_0(x) + c_2K_0(x) - q$$

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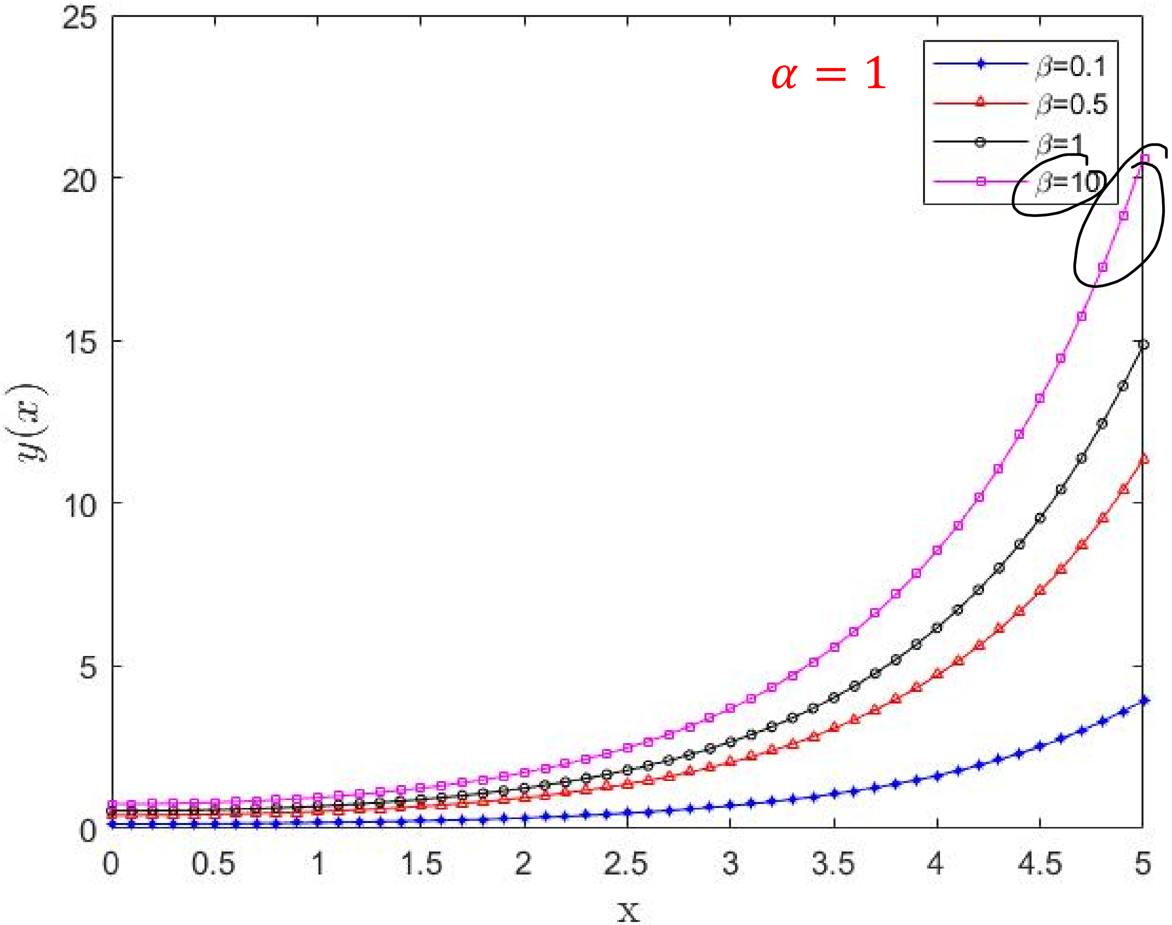
Dimensionless Concentration Profile



mass transfer

$$\beta = \frac{k_c}{\sqrt{k a D}}$$

diffusion rxn

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Catalyst Efficiency FYI

with resistance.

Define υ: actual reaction rate in the catalyst

$$\upsilon = \left(D\frac{dC_A}{dr}2\pi rL\right)_{r=R} = \frac{2\pi RLDC_{A,0}}{L_d}\boxed{\left(\frac{dy}{dx}\right)_{x=\alpha}}$$

$y'(\alpha)$

Define γ: reaction rate without diffusion resistance in the catalyst

$$\gamma = kC_{A,0}a\pi R^2L$$

$$y'(\alpha) = \frac{\beta I_1(\alpha)}{\beta I_0(\alpha)+I_1(\alpha)} \implies \upsilon = \boxed{\frac{2\pi RLDC_{A,0}}{L_d}}\frac{\beta I_1(\alpha)}{\beta I_0(\alpha)+I_1(\alpha)}$$

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Catalyst Efficiency FYI

$$
\begin{aligned}
\eta &= \frac{v}{\gamma} \\
&= \frac{\frac{2\cancel{\pi} R \cancel{L} D \cancel{C_{A,0}}}{L_d} \cdot \frac{\beta I_1(\alpha)}{\beta I_0(\alpha)+I_1(\alpha)}}{k \cancel{C_{A,0}} a \cancel{\pi} R^2 \cancel{L}} \\
&= \frac{2D\beta I_1(\alpha)}{akRL_d(\beta I_0(\alpha) + I_1(\alpha))} \\
&= \frac{2}{R\sqrt{\frac{ka}{D}}} \frac{\beta I_1(\alpha)}{\beta I_0(\alpha) + I_1(\alpha)} \\
&= \frac{2}{\alpha} \frac{\beta I_1(\alpha)}{\beta I_0(\alpha) + I_1(\alpha)}
\end{aligned}
$$

Catalyst Efficiency FYI

Assume $\beta \gg 1$

$$\beta = \frac{k_c}{\sqrt{kaD}}$$

$$\eta = \frac{2}{\alpha}\frac{\beta I_1(\alpha)}{\beta I_0(\alpha) + \cancel{I_1(\alpha)}} \sim \frac{2}{\alpha}\frac{I_1(\alpha)}{I_0(\alpha)}$$

$$\underset{\alpha \to 0}{I_1(\alpha)} = \frac{\alpha^1}{2\underbrace{\Gamma(1+1)}_{1!}} = \frac{\alpha}{2}$$

For limiting small α :

$$\lim_{\alpha \to 0} \eta \sim \frac{2}{\alpha}\frac{I_1(0)}{I_0(0)} \sim \frac{2}{\alpha}\frac{\frac{\alpha}{2}}{1} = 1$$

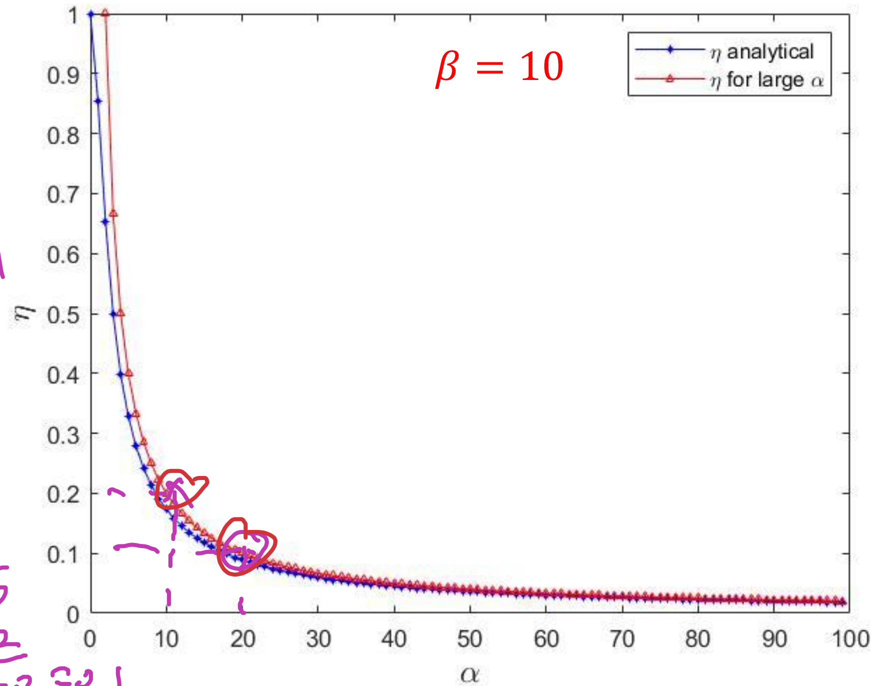
$$\frac{\alpha^0}{2^0\,\Gamma(1)} = \frac{1}{1 \cdot 1} = 1$$

For limiting large α :

$$\lim_{\alpha \to \infty} \eta \sim \frac{2}{\alpha}\frac{\frac{e^\alpha}{\sqrt{2\pi\alpha}}}{\frac{e^\alpha}{\sqrt{2\pi\alpha}}} \sim \frac{2}{\alpha}$$

$\alpha = 10,\ y = \frac{2}{10}$

$\alpha = 20,\ y = \frac{2}{20} = 0.1$



Properties of Bessel Functions

Function	Small x	Large x
$J_p(x)$	$\frac{x^p}{2^p\Gamma(p+1)}$	$\sqrt{\frac{2}{\pi x}}\cos\left(x-\frac{\pi+p\pi}{4}\right)$
$Y_0(x)$	$\frac{2}{\pi}\ln x$	$\sqrt{\frac{2}{\pi x}}\sin\left(x-\frac{\pi+p\pi}{4}\right)$
$Y_p(x), p\neq 0$	$\frac{-2^p(p-1)!}{\pi x^p}$	$\sqrt{\frac{2}{\pi x}}\sin\left(x-\frac{\pi+p\pi}{4}\right)$
$I_p(x)$	$\frac{x^p}{2^p\Gamma(p+1)}$	$\frac{e^x}{\sqrt{2\pi x}}$
$K_0(x)$	$-\ln x$	$\sqrt{\frac{\pi}{2x}}e^{-x}$
$K_p(x), p\neq 0$	$\frac{2^{p-1}(p-1)!}{x^p}$	$\sqrt{\frac{\pi}{2x}}e^{-x}$

$$p=\frac{1}{2}:$$

$$J_{\frac{1}{2}}(x)=\sqrt{\frac{2}{\pi x}}\sin(x)$$

$$J_{-\frac{1}{2}}(x)=\sqrt{\frac{2}{\pi x}}\cos(x)$$

$$I_{\frac{1}{2}}(x)=\sqrt{\frac{2}{\pi x}}\sinh(x)$$

$$I_{-\frac{1}{2}}(x)=\sqrt{\frac{2}{\pi x}}\cosh(x)$$

$$J_0'(x)=-J_1(x) \qquad J_p(0)=\begin{cases}1 & p=0\\ 0 & p=1,2,\ldots\end{cases}$$

$$Y_0'(x)=-Y_1(x) \qquad Y_p(0)=-\infty$$

$$I_0'(x)=I_1(x) \qquad I_p(0)=\begin{cases}1 & p=0\\ 0 & p=1,2,\ldots\end{cases} \quad \& \; Y_p(\infty)=\infty$$

$$K_0'(x)=-K_1(x) \qquad K_p(0)=\infty \;\&\; K_p(\infty)=0$$

Properties of Bessel Functions

$$
\begin{aligned}
&\text{When } Z_n = J_n \text{ or } Y_n: \\
&\frac{2n}{x} Z_n(x) = Z_{n+1}(x) + Z_{n-1}(x) \\
&\frac{d}{dx}\left[Z_n(x)\right] = Z_{n-1}(x) - \frac{n}{x} Z_n(x) = \frac{n}{x} Z_n(x) - Z_{n+1}(x) \\
&\frac{d}{dx}\left[x^n Z_n(x)\right] = x^n Z_{n-1}(x); \quad Z_0'(x) = -Z_1(x)
\end{aligned}
$$

$$
\begin{aligned}
&\text{When } Z_n = I_n: \\
&\frac{2n}{x} I_n(x) = I_{n-1}(x) - I_{n+1}(x) \\
&\frac{d}{dx}\left[I_n(x)\right] = I_{n-1}(x) - \frac{n}{x} I_n(x) = \frac{n}{x} I_n(x) + I_{n+1}(x) \\
&\frac{d}{dx}\left[x^n I_n(x)\right] = x^n I_{n-1}(x); \quad I_0'(x) = I_1(x)
\end{aligned}
$$

$$
\begin{aligned}
&\text{When } Z_n = K_n: \\
&\frac{2n}{x} K_n(x) = K_{n+1}(x) - K_{n-1}(x) \\
&\frac{d}{dx}\left[K_n(x)\right] = -K_{n-1}(x) - \frac{n}{x} K_n(x) = \frac{n}{x} K_n(x) - K_{n+1}(x) \\
&\frac{d}{dx}\left[x^n K_n(x)\right] = -x^n K_{n-1}(x); \quad K_0'(x) = -K_1(x)
\end{aligned}
$$

Properties of Bessel Functions

When $Z_n = J_n$ or Y_n

$$\int uZ_0^2(u)du = \frac{u^2}{2}\left[Z_0^2(u) + Z_1^2(u)\right] + C$$

$$\int uZ_1^2(u)du = \frac{u^2}{2}\left[Z_0^2(u) + Z_1^2(u)\right] - uZ_0(u)Z_1(u) + C$$

$$\int u^2 Z_0(u)Z_1(u)du = \frac{u^2}{2}Z_1^2(u) + C$$

When $Z_n = I_n$

$$\int uI_0^2(u)du = \frac{u^2}{2}\left[I_0^2(u) - I_1^2(u)\right] + C$$

$$\int uI_1^2(u)du = uI_0(u)I_1(u) - \frac{u^2}{2}\left[I_0^2(u) - I_1^2(u)\right] + C$$

$$\int u^2 I_0(u)I_1(u)du = \frac{u^2}{2}I_1^2(u) + C$$

When $Z_n = K_n$

$$\int uK_0^2(u)du = \frac{u^2}{2}\left[K_0^2(u) - K_1^2(u)\right] + C$$

$$\int uK_1^2(u)du = -uK_0(u)K_1(u) - \frac{u^2}{2}\left[K_0^2(u) - K_1^2(u)\right] + C$$

$$\int u^2 K_0(u)K_1(u)du = -\frac{u^2}{2}K_1^2(u) + C$$

Generalized Bessel ODE

$$x^2y'' + x(a + 2bx^r)y' + \left[c + dx^{2s} - b(1 - a - r)x^r + b^2x^{2r}\right] y = 0; \quad \text{for } r, s \neq 0$$

with general solution in the form

$$y = x^{\frac{1-a}{2}} e^{-\frac{bx^r}{r}} \left[AZ_p\left(\frac{\sqrt{|d|}}{s}x^s\right) + BZ_{-p}\left(\frac{\sqrt{|d|}}{s}x^s\right)\right] \text{ where } p = \frac{1}{s}\sqrt{\left(\frac{1-a}{2}\right)^2 - c}$$

1. If $\frac{\sqrt{d}}{s}$ is *real* and p is *not* an integer (or zero), then Z_p denotes J_p and Z_{-p} denotes J_{-p}
2. If $\frac{\sqrt{d}}{s}$ is *real* and p is zero or an integer k, then Z_p denotes J_k and Z_{-p} denotes Y_k
3. If $\frac{\sqrt{d}}{s}$ is *imaginary* and p is *not* zero or an integer, then Z_p denotes I_p and Z_{-p} denotes I_{-p}
4. If $\frac{\sqrt{d}}{s}$ is *imaginary* and p is zero or an integer k, then Z_p denotes I_k and Z_{-p} denotes K_k

HW1 Q2: Solve for the dimensionless concentration

2. A water stream with a constant volumetric flow rate Q is fed into a pipe of radius R with a permeable wall, as illustrated below. The pipe is embedded in a porous medium that extends infinitely in the radial direction but has a finite height L. The water contains a dissolved reagent at an initial concentration C_F which undergoes a first-order reaction on the surface of the solid particles within the porous medium. The reaction rate is given by $r_s = kC\left(\frac{\text{mole}}{\text{cm}^2\text{s}}\right)$. The molar flux of the reagent N is expressed as

$$N = Cv - D\frac{dC}{dr}$$

where v is the superficial velocity and D is the effective diffusivity.
(a) Using the shell balance technique, show that the steady-state concentration distribution is governed by the equation:

$$r^2\frac{d^2C}{dr^2} + r\left(1 - \frac{rv}{D}\right)\frac{dC}{dr} - \frac{ak}{D}r^2C = 0$$

where a is the specific interfacial area of the porous solid $\left(\frac{\text{cm}^2}{\text{cm}^3}\right)$. Clearly state any assumptions made in the derivation. [*Hint: For dilute solutions, mass continuity requires* $\frac{d}{dr}(rv) = 0$].
(b) Identify the appropriate boundary conditions for the governing equation. *Hint: Consider the two physical boundaries in the system.*
(c) Nondimensionalize the governing equation and its boundary conditions.

HW1 Q2: Solve for the dimensionless concentration

(c) From mass continuity condition $\frac{d(rv)}{dr} = 0$, rv is constant and hence $rv = Rv(R)$. With (12) to obtain $rv = \frac{Q}{2\pi L}$ and then substitute this into equation (10),

$$r^2 \frac{d^2C}{dr^2} + r\left(1 - \frac{Q}{2\pi LD}\right)\frac{dC}{dr} - \frac{ak}{D} r^2 C = 0 \tag{15}$$

Let $\widetilde{C} = \frac{C}{C_F}$ and $\widetilde{r} = \frac{r}{R}$, the equation (15) can be transformed into its dimensionless form as

$$\widetilde{r}^2 \frac{d^2\widetilde{C}}{d\widetilde{r}^2} + \widetilde{r}\,(1-\theta)\frac{d\widetilde{C}}{d\widetilde{r}} - \phi\widetilde{r}^2\widetilde{C} = 0 \quad \text{Solve.} \tag{16}$$

where $\theta = \frac{Q}{2\pi LD}$ and $\phi = \frac{akR^2}{D}$. The dimensionless boundary conditons are

$$\widetilde{C}(1) - \frac{2\pi LD}{Q}\frac{d\widetilde{C}}{d\widetilde{r}}\bigg|_{\widetilde{r}=1} = 1 \tag{17}$$

$$\widetilde{C}(\widetilde{r} \to \infty) = 0 \text{ or } \frac{d\widetilde{C}}{d\widetilde{r}}\bigg|_{\widetilde{r}\to\infty} = 0 \tag{18}$$

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HW1 Q2: Solve for the dimensionless concentration

Bonus Qn for class

Announcements

Week 5 Homework

- Due Date: Week 6 class – 2:00 PM, March 4, 2026
- Required group work and intergroup discussions are encouraged

Week 6 Class

- Lab Session: Please bring your laptop with MATLAB installed
- Mid-Term Project Consultation

Week 7 Individual Meetings

- Date: Monday, March 9, 2026, Time: **1:45 PM – 4:15 PM**
- Date: Wednesday, March 11, 2026, Time: **10:00 AM – 12:30 PM**
- Action Required: Please indicate your time slot using this link

https://docs.google.com/spreadsheets/d/10sE6iWVvFoSE9Oo5DYvNaOhwu1nqeNufKXq8NkKbHRs/edit?gid=0#gid=0